Anirudh Chandan

Product Engineer · TypeScript / Node.js · PostgreSQL · Regulated-data platforms

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SUMMARY

Engineer (2.5+ years incl. freelance) who builds **platforms for regulated, audited work**: a healthcare EHR used by 2,300+ clinics, and a document-workflow platform with **immutable versions, maker-checker sign-off, rule-defined validation and full audit trails** — configurable by the customer rather than hard-coded. TypeScript/Node.js and PostgreSQL first, with Python and React where needed; GCP and AWS in production. First-principles builder: wrote **PyDB, a transactional storage engine from scratch** (B-Trees, write-ahead log, crash recovery). 450+ LeetCode.

EXPERIENCE

Independent Software Contractor *Aug 2026 – Present*
Backend / Full-Stack Engineer (freelance) *Remote · India / Australia*

- Designed and shipped a **multi-tenant document workflow platform** (FastAPI, PostgreSQL with row-level security, Celery): **immutable document versions, maker-checker approval with recorded overrides, per-check audit trace**, and **customer-editable YAML rule packs** (cross-document checks such as bill-of-lading weights vs manifest ±0.5%) that apply without a deploy. Built as a platform, not a product — a second industry pack (procure-to-pay, 3-way match) ran with zero engine changes. 400+ tests.
- Delivered a **Node.js operations dashboard** for an on-site AI-CCTV deployment: webhook ingestion, compliance timers, role-based access, printable reports — installed it on site myself, **in production since Sep 2026**; I am the person who hears when it breaks.

Docplix *Nov 2025 – Present*
Software Engineer *Remote*

- Built and led 30+ Node.js/Sequelize APIs for a **regulated healthcare EHR serving 2,300+ clinics**, one of **4 engineers**; ran a **zero-downtime v1→v2 migration with 100% data integrity** on live patient data.
- Designed **optimistic concurrency control and distributed Redis locks** for the central sync service, eliminating race conditions behind concurrent-write data mismatches.

Genpact *Feb 2024 – Oct 2025*
Software Engineer (promoted from Intern) *Bengaluru, India*

- Engineered **idempotent, exactly-once processing** for a Kafka tax-ingestion pipeline using database constraints, guaranteeing correctness across partitions during network failures.
- Built resilient Python (Flask) microservices with automated retries and dead-letter queues, serving 24 modules with **zero data loss**.

PROJECTS

PyDB — Database Storage Engine Built From Scratch. A disk-based transactional storage engine in pure Python: its own **pager and 8 KB page cache**, raw binary serialization, **clustered and secondary B-Tree indexes**, and a **write-ahead log** with redo recovery and atomic checkpoints. Recovery is tested by **SIGKILLing a writer mid-workload**; CI on every push. (live demo · code)

ProjAuto — Multi-tenant SaaS ERP (20+ live tenants, 120+ services). Top contributor (711 of 1,100 commits), full-stack: migrated a legacy Java monolith to **Spring Boot + React**, eliminated N+1 queries across 53 endpoints (**15.6s → 2.4s**), built a distributed token-bucket rate limiter, and closed account-takeover and cross-tenant IDOR security holes.

SKILLS

Core: TypeScript, Node.js (Express, Sequelize), PostgreSQL, REST API design, event-driven services, WebSockets; Python (FastAPI, Flask), Java (Spring Boot), React (ProjAuto)

Data: PostgreSQL (row-level security, indexing, migrations on live data), MySQL, Redis (caching, Pub/Sub, distributed locks), Kafka, BullMQ, Celery, database internals (B-Trees, WAL, ACID)

Cloud: GCP (Cloud Functions, serverless), AWS, Docker, Linux, GitHub Actions CI/CD

Practices: end-to-end ownership (spec → architecture → ship → on-call), audit trails & versioned configuration, concurrency control, testing (Vitest, pytest), distributed tracing, AI-assisted development (Claude Code) with accountability for what lands

Foundations: Data Structures & Algorithms (450+ LeetCode), OOP, SQL

EDUCATION & ACHIEVEMENTS

Graphic Era University, Dehradun *2020 – 2024*

B.Tech, Computer Science & Engineering — CGPA 9.11

Technical blog on systems internals (anirudh-chandan.vercel.app/blog).